Acceleration

LEARNING TARGET · R4 I can use velocity–time data to find how fast velocity changes.

SECTION A KEY VOCABULARY

Acceleration	The rate velocity changes: how much velocity is gained or lost each second.
Speeding up	v and a have the same sign. Gaps between equal-time dots grow.
Slowing down	v and a have opposite signs. Braking is acceleration too.
Turning	A direction change is a velocity change, even at steady speed.

SECTION B KEY EQUATIONS + UNITS

$$a = \Delta v \div \Delta t$$

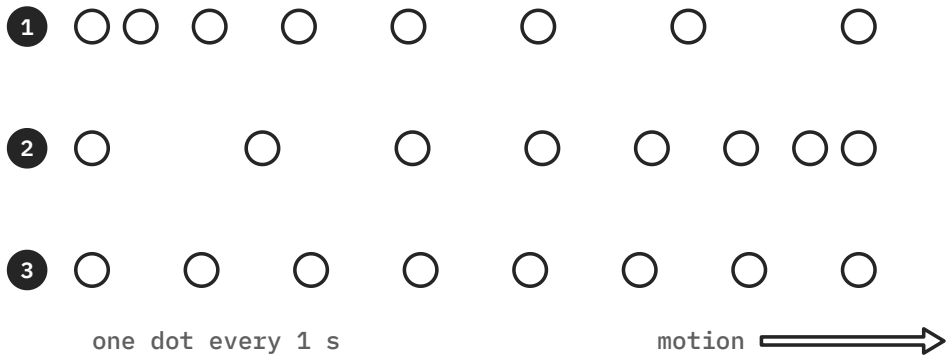
$$v = v_0 + a\,t$$

a m/s² ("m/s, every second")

g ≈ 9.8 m/s² (free fall)

SECTION C COLOR IT · READ THE GAPS

Equal time between dots.



☐ **1 · GREEN** Speeding up: gaps grow

☐ **2 · RED** Slowing down: gaps shrink

☐ **3 · YELLOW** Steady: a = 0

SECTION D WORKED EXAMPLE

A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration.

GIVEN	EQUATION	WORK	ANSWER
$v_i = 0$ m/s $v_f = 8$ m/s $\Delta t = 4$ s	$a = \Delta v \div \Delta t$	$\Delta v = 8 - 0 = 8$ m/s $a = 8 \div 4$	**2 m/s²** gains 2 m/s each second

SECTION E THE 2026-XJ THREAD

DAY 7 OF 22 · NEW TOOL: ACCELERATION

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else